13.5 COMMUNALITIES

The communalities are computed by taking the sum of squares of the loadings for the i[th] variable on the m common factors. This can be shown as below

Variance due to the specific factor is often called the uniqueness or specific variance. It is denoted by h_i^2, we have from equation (13.2)

$$\sigma_{ii} = l_{i1}^2 + l_{i2}^2 + \cdots + l_{im}^2 + \Psi_i$$

or $$\sigma_{ii} = h_i^2 + \Psi_i \; ; \qquad i = 1, 2, \ldots, p \qquad (13.5)$$

here, $$h_i^2 = l_{i1}^2 + l_{i2}^2 + \cdots + l_{im}^2$$

Now, to understand the computation of communalities in a simple way, let us have the table of three factor loadings. The following dataset involves the Places Rated Almanac (Boyer and Savageau) rates 329 communities according to nine criteria:

These data are only recorded for the first three factors only. It should also note that these factor loadings are the correlations between the factors and the variables. For example, the correlation between the Arts and the first factor is about 86%. Similarly the correlation between climate and that factor is only about 28%.

	Factor		
Variable	**1**	**2**	**3**
Climate	0.286	0.076	0.841
Housing	0.698	0.153	0.084
Health	0.744	-0.410	-0.020
Crime	0.471	0.522	0.135
Transportation	0.681	-0.156	-0.148
Education	0.498	-0.498	-0.253
Arts	0.861	-0.115	0.011
Recreation	0.642	0.322	0.044
Economics	0.298	0.595	-0.533

For example, to compute the communality for climate, the first variable which is given in the above table, we square the factor loadings for climate and then add the results:

$$h_1^2 = 0.286^2 + 0.076^2 + 0.841^2 = 0.795$$

In the similar way we can obtain the communalities for all the 9 variables and the total communality is obtained as 5.617. Therefore, all the communalities computed are placed into a table shown below:

Variable	Communality
Climate	0.795
Housing	0.518
Health	0.722
Crime	0.512
Transportation	0.510
Education	0.561
Arts	0.754
Recreation	0.517
Economics	0.728
Total	**5.617**

These values are as similar as multiple R^2 values for regression models predicting the variables of interest from the 3 factors. The communality for a given variable can be interpreted as the proportion of variation in that variable explained by the three factors. In other words, if we perform multiple regression of climate against the three common factors, we obtain an R^2= 0.795, indicating that about 79% of the variation in climate is explained by the factor model. The results suggest that the factor analysis does the best job of explaining variation in climate, the arts, economics, and health.

One assessment of how well this model is doing can be obtained from the communalities. What you want to see is values that are close to one. This would indicate that the model explains most of the variation for those variables. In this case, the model does better for some variables than it does for others. The model explains Climate the best, and is not bad for other variables such as Economics, Health and the Arts. However, for other variables such as Crime, Recreation, Transportation and Housing the model does not do a good job, explaining only about half of the variation. If you take all of the communality values and add them up you can get a total communality value:

$$\sum_{i=1}^{p} h_i^2 = \sum_{i=1}^{m} \lambda_i$$

Here, the total communality is 5.617. The proportion of the total variation explained by the three factors is

$$\frac{5.617}{9} = 0.624$$

This gives us the percentage of variation explained in our model. This might be looked at as an overall assessment of the performance of the model. However, this percentage is the same as the proportion of variation explained by the first three eigenvalues, obtained earlier. The individual communalities tell how well the model is working for the individual variables, and the total communality gives an overall assessment of performance. These are two different assessments that can be used.

Since the data are standardized in this case, the variance for standardized data is going to be equal to one. Then the specific variances can be computed by subtracting the communality from the variance as expressed below:

$$\Psi_i = 1 - h_i^2$$

Recall, that the data were standardized before analysis, so the variances of the standardized variables are all equal to one. For example, the specific variance for Climate is computed as follows:

$$\Psi_1 = 1 - 0.795 = 0.205$$

The specific variance for housing is Ψ_2 =0.482 and the other values of specific variances can be calculated accordingly.

Source: http://onlinecourses.science.psu.edu

EXAMPLE 1 Calculating Correlations from Factors

Factor analysis aims to explain the correlations among the observed variables in terms of small number of factors. To estimate the success of factor solution, we reproduce the original correlation matrix by using the loadings on the common factors and see how large a discrepancy is there between the original matrix and reproduced correlation matrix. The greater the discrepancy, the less successful the factor solution in preserving the information in the original correlation matrix. And, lesser the discrepancy, more successful the factor solution in preserving the information in original correlation matrix.

In order to calculate the correlations from the factor solution the process is quite simple when the factors are uncorrelated. The correlation between any two variables X_1 and X_2 is obtained by adding products of coefficients for these variables (X_1 and X_2) across all the common factors. Let us say for the three factor solution F_1, F_2 and F_3 the quantity would be $(l_{11} \times l_{21}) + (l_{12} \times l_{22}) + (l_{13} \times l_{23})$. This process will become clearer in the description of the hypothetical two-factor solution based on five observed variables as follows:

A Hypothetical Solution

Variables	**Loadings/ Correlations**		**Communality**	**Reproduced Correlations**					
	Factor 1	**Factor 2**			**X_1**	**X_2**	**X_3**	**X_4**	**X_5**
X_1	.5	.2	.29	X_1					
X_2	.6	.3	.45	X_2	.36				
X_3	.7	.4	.65	X_3	.43	.54			
X_4	.3	.6	.45	X_4	.27	.36	.45		
X_5	.4	.7	.65	X_5	.34	.45	.56	.54	
	$\sum x^2 = 1.35$	$\sum x^2 = 1.14$							

The Coefficients: According to the above solution,

$X_1 = 0.5F_1 + 0.2F_2$

$X_2 = 0.6F_1 + 0.3F_2$

$\vdots$

$X_5 = 0.4F_1 + 0.7F_2$

As well as being weights, the coefficients above show that the correlation between X_1 and F_1 is .50, and that of between X_1 and F_2 is .20, and so on.

Variance Accounted For: The quantities at the bottom of each factor column are the sums of the squared loadings for that factor, and show how much of the total variance of the observed variables is accounted for by that factor. For Factor 1, the quantity is $.5^2 + .6^2 + .7^2 + .3^2 + .4^2 = 1.35$. Because in the factor analyses discussed here, the total amount of variance is equal to the number of observed variables (the variables are standardized, so each has a variance of one), the total variation here is five, so that Factor 1 accounts for $(1.35/5)\times100 = 27\%$ of the variance.

The quantities in the communality column show the proportion of the variance of each variable accounted for by the common factors. For X_1 this quantity is $.5^2 + .2^2 = 0.29$, for X_2 it is $.6^2 + .3^2 = 0.45$, and so on.

Reproducing the Correlations: The correlation between variables X_1 and X_2 as derived from the factor solution is equal to (.7 X .8) + (.2 X .3) = 0.36, while the correlation between variables X_3 and X_5 is equal to (.7 X .4) + (.4 X .7) = .56. These values are shown in right-hand side of the above table.

EXAMPLE 2 To verify the relation $\Sigma = \mathrm{L}\,\mathrm{L}' + \Psi$ for two factors:

Consider the following covariance matrix

$$\Sigma = \begin{bmatrix} 19 & 30 & 2 & 12 \\ 30 & 57 & 5 & 23 \\ 2 & 5 & 38 & 47 \\ 12 & 23 & 47 & 68 \end{bmatrix}$$

The equality equation of the model is

$$\begin{bmatrix} 19 & 30 & 2 & 12 \\ 30 & 57 & 5 & 23 \\ 2 & 5 & 38 & 47 \\ 12 & 23 & 47 & 68 \end{bmatrix} = \begin{bmatrix} 4 & 1 \\ 7 & 2 \\ -1 & 6 \\ 1 & 8 \end{bmatrix} \begin{bmatrix} 4 & 7 & -1 & 1 \\ 1 & 2 & 6 & 8 \end{bmatrix} + \begin{bmatrix} 2 & 0 & 0 & 0 \\ 0 & 4 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 3 \end{bmatrix}$$

or

$$\Sigma = \mathrm{L}\,\mathrm{L}' + \Psi$$

may be verified by the matrix algebra. Therefore, Σ has the structure produced by an m= 2 orthogonal factor model. Since

$$\mathrm{L} = \begin{bmatrix} l_{11} & l_{12} \\ l_{21} & l_{22} \\ l_{31} & l_{32} \\ l_{41} & l_{42} \end{bmatrix} = \begin{bmatrix} 4 & 1 \\ 7 & 2 \\ -1 & 6 \\ 1 & 8 \end{bmatrix}$$

$$\Psi = \begin{bmatrix} \psi_1 & 0 & 0 & 0 \\ 0 & \psi_2 & 0 & 0 \\ 0 & 0 & \psi_3 & 0 \\ 0 & 0 & 0 & \psi_4 \end{bmatrix} = \begin{bmatrix} 2 & 0 & 0 & 0 \\ 0 & 4 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 3 \end{bmatrix}$$

The communality of X_i can be obtained by using equation (13.5), we have

$$h_i^2 = l_{11}^2 + l_{12}^2 = 4^2 + 1^2 = 17$$

and the variance of X_i can be decomposed as

$$\sigma_{11} = (l_{11}^2 + l_{12}^2) + \psi_1 = h_1^2 + \psi_1$$

or

$$19 = 4^2 + 1^2 + 2 = 17 + 2$$

i.e. variance = communality+ specific variance

Similarly, we can find out the results for the other variables also.

Check Your Progress 1

1) What do you mean by a factor in the context of factor analysis?

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2) Show that the covariance matrix

$$\Sigma = \begin{bmatrix} 1.0 & .63 & .45 \\ .63 & 1.0 & .35 \\ .45 & .35 & 1.0 \end{bmatrix}$$

For the p=3 standardized random variables Z_1, Z_2 and Z_3 can be generated by the m=1 factor model:

$$Z_1 = .9F_1 + \varepsilon_1$$

$$Z_2 = .7F_1 + \varepsilon_2$$

$$Z_3 = .5F_1 + \varepsilon_3$$

where, $\mathrm{Var}\,(F_1) = 1, \quad \mathrm{Cov}\,(\varepsilon, F_1) = 0$

and $\Psi = \mathrm{Cov}\,(\varepsilon) = \begin{bmatrix} .19 & 0 & 0 \\ 0 & .51 & 0 \\ 0 & 0 & .75 \end{bmatrix}$

Formulate the factor model for the given problem and also calculate

(a) Communalities h_i^2; i= 1, 2, 3 and interpret the result.

(b) Cor (Z_i, F_i) for i= 1, 2, 3. Which variable carry the greatest weight in naming the common factor? And why?

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3) Let the factor model with p=2 and m=1. Show that

$$\sigma_{11} = l_{11}^2 + \psi_1$$

$$\sigma_{12} = \sigma_{21} = l_{11}l_{21}$$

$$\sigma_{22} = l_{21}^2 + \psi_2$$

for given σ_{11}, σ_{22} and σ_{12}, there is an infinite choices for L and Ψ i.e., the factor model need not be unique.

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4) Consider an m=1 factor model for the population with covariance matrix

$$\Sigma = \begin{bmatrix} 1 & .4 & .9 \\ .4 & 1 & .7 \\ .9 & .7 & 1 \end{bmatrix}$$

Show that there is a unique choice of L and Ψ with $\Sigma = L\,L' + \Psi$, but that $\psi_3 < 0$, so the choice is not admissible i.e., the solution is unique but improper.

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13.6 METHODS OF ESTIMATION

For the observations $x_1, x_2, \ldots\ldots, x_n$ on p generally correlated variable, does the factor model of equation (13.1) with a small number of factors, effectively represent the data? The solution of such kind of statistical problems may be tackled by using the covariance relationships. The sample covariance matrix S, is an unbiased estimate of the unknown population covariance matrix Σ. If Σ appears to deviate significantly from a diagonal matrix, then a factor model can be entertained, and the initial problem is one of estimating the factor loadings l_{ij} and specific variances ψ_i.

In this section, we shall consider two of the most popular methods of parameter estimation, the principal component (and the related principal factor) method and the maximum likelihood method. The solution from either method can be rotated in order to simplify the interpretation of factors.

13.6.1 Principal Component Method

A vector of observations for the X_i variable may be defined as

$$X = \begin{bmatrix} X_1 \\ X_2 \\ \vdots \\ X_p \end{bmatrix}$$

The sample covariance matrix is denoted by S and is expressed as

$$S = \frac{1}{n-1}\sum_{i=1}^{n}(X_i - \bar{X})\,(X_i - \bar{X})'$$

We have p eigenvalues for the covariance matrix S as well as corresponding eigenvectors for this matrix. Let the eigenvalues of the covariance matrix S are $\hat{\lambda}_1, \hat{\lambda}_2, \ldots \ldots . \hat{\lambda}_p$ and eigenvectors of the covariance matrix S are $\hat{e}_1, \hat{e}_2, \ldots \ldots . \hat{e}_p$. The covariance matrix can be re-expressed in the following form as a function of the eigenvalues and the eigenvectors:

Spectral Decomposition of Σ: The spectral decomposition allows us to express the inverse of a square matrix in terms of its eigenvalues and eigenvectors, and this leads to a useful square-root matrix. Let the population covariance matrix Σ have eigenvalues and eigenvectors pairs (λ_i, e_i) with $\lambda_1 \geq \lambda_2 \geq \cdots \geq \lambda_p \geq 0$, then

$$\Sigma = \lambda_1 e_1 e_1' + \lambda_2 e_2 e_2' + \cdots + \lambda_p e_p e_p' = \sum_{i=1}^{p} \lambda_i e_i e_i'$$

$$= \left[\sqrt{\lambda_1}e_1 \ \vdots\ \sqrt{\lambda_2}e_2 \ \vdots\ \cdots\cdots \ \vdots\ \sqrt{\lambda_p}e_p\right]\begin{bmatrix}\sqrt{\lambda_1}e_1' \\ \cdots\cdots \\ \sqrt{\lambda_2}e_2' \\ \cdots\cdots \\ \vdots \\ \cdots\cdots \\ \sqrt{\lambda_p}e_p'\end{bmatrix} \qquad (13.6)$$

This fits the prescribed covariance structure for the factor analysis model having as many factors as variables i.e., m= p and specific variances $\Psi_i = 0$ for all i. The loading matrix has jth column given by $\sqrt{\lambda_j}e_j$. Then we can write

$$\Sigma = L\,L + 0 = L\,L \qquad (13.7)$$

Apart from the scale factor $\sqrt{\lambda_j}$, the factor loadings on the jth factor are the coefficients for the jth principal component of the population. The factor analysis representation of Σ in equation (13.7) is exact, it is not particularly useful. It employs as many common factors as there are variables and does not allow any variation in the specific factors ε in the main model of equation (13.1). But we prefer models that explain the covariance structure in terms of just a few common factors. One approach when the last $p - m$ eigenvalues are small is to neglect the contribution of $\lambda_{m+1}e_{m+1}e_{m+1}' + \cdots + \lambda_p e_p e_p'$ to Σ in eq (13.6), we obtain the approximation

$$\Sigma = \left[\sqrt{\lambda_1}e_1 \ \vdots\ \sqrt{\lambda_2}e_2 \ \vdots\ \cdots\cdots \ \vdots\ \sqrt{\lambda_m}e_m\right]\begin{bmatrix}\sqrt{\lambda_1}e_1' \\ \cdots\cdots \\ \sqrt{\lambda_2}e_2' \\ \cdots\cdots \\ \vdots \\ \cdots\cdots \\ \sqrt{\lambda_m}e_m'\end{bmatrix} = L\,L \qquad (13.8)$$

The approximation shown in eq (13.8) assumes that the specific factors ε in eq (13.1) are of minor importance and can also be ignored in the factoring of Σ.

Allowing for specific factors, we find that the approximation becomes

$$\Sigma = \mathrm{L\,L'} + \Psi$$

$$= \left[\sqrt{\lambda_1}\mathrm{e}_1 \;\vdots\; \sqrt{\lambda_2}\mathrm{e}_2 \;\vdots\; \cdots\cdots \;\vdots\; \sqrt{\lambda_m}\mathrm{e}_m\right] \begin{bmatrix} \sqrt{\lambda_1}\mathrm{e}_1' \\ \cdots\cdots \\ \sqrt{\lambda_2}\mathrm{e}_2' \\ \cdots\cdots \\ \vdots \\ \cdots\cdots \\ \sqrt{\lambda_m}\mathrm{e}_m' \end{bmatrix} + \begin{bmatrix} \psi_1 & 0 & \cdots & 0 \\ 0 & \psi_2 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & \psi_p \end{bmatrix} \quad (13.9)$$

where

$$\psi_i = \sigma_{ii} - \sum_{j=1}^{m} l_{ij}^2\,; \qquad i = 1, 2, \ldots\ldots, p$$

This yields the following estimator for the factor loadings

$$\hat{l}_{ij} = \hat{e}_{ij}\sqrt{\hat{\lambda}_i}$$

When the units of the variables are not appropriate, it is better to use the standardized variables. If the standardized measurements are used, we replace sample correlation matrix S by R. This in turn suggests that the specific variances, which are the diagonal elements of the matrix Ψ, can be estimated by using the expression:

$$\hat{\psi}_i = S_{ii} - \sum_{j=1}^{m} \hat{l}_{ij}^2\,; \qquad i = 1, 2, \ldots\ldots, p \qquad (13.10)$$

Here, the estimated specific variances are equal to the sample variance for the i^{th} variable minus the sum of the squares of factor loadings (i.e the communality).

The equation (13.10), when applied to the sample covariance matrix S or the sample correlation matrix R, is known as the principal component solution.

13.6.2 Maximum Likelihood Method

Maximum likelihood estimation requires that the data are sampled from a multivariate normal distribution. But if the data have been collected on a likert scale, which is most often the case in the social sciences, these kinds of data cannot really be normally distributed. This is the main drawback of this method. Therefore, using the maximum likelihood estimation method we must assume that the data are independently sampled from a multivariate normal distribution with mean vector μ and the covariance matrix Σ that takes the particular form

$$\Sigma = \mathrm{L\,L'} + \Psi$$

where L is the matrix of factor loadings and Ψ is the diagonal matrix of specific variances.

Again, let us assume that the p×1 vectors $X_1, X_2, \ldots\ldots X_n$ represents a random sample which is drawn from a multivariate normal population $N_p(\mu, \Sigma)$, then the joint probability function or the likelihood function is given by

$$l(\mu, \Sigma) = \prod_{j=1}^{n} \left\{ \frac{1}{(2\pi)^{p/2}|\Sigma|^{1/2}} \; e^{-(x_j-\mu)'\Sigma^{-1}(x_j-\mu)/2} \right\}$$

$$= \frac{1}{(2\pi)^{np/2}|\Sigma|^{n/2}} \; e^{-\sum_{j=1}^{n}(x_j-\mu)'\Sigma^{-1}(x_j-\mu)/2} \qquad (13.11)$$

Maximum likelihood estimation involves estimating the mean, the matrix of factor loadings and the specific variances.

The maximum likelihood estimator for the mean vector μ , the factor loadings L and the specific variances Ψ are obtained by finding the values of $\hat{\mu}$, $\hat{L}$ and $\hat{\Psi}$ that maximizes by taking logarithm of the likelihood function in equation (13.11) which is given by the following expression:

$$\text{Log } l(\mu, L, \Psi) = -\frac{np}{2}\log 2\pi$$

$$-\frac{n}{2}\log\left|LL' + \Psi\right| - \frac{1}{2}\sum_{j=1}^{n}\left(x_j - \mu\right)'(LL' + \Psi - 1)\left(x_j - \mu\right)$$

(13.12)

The log of the likelihood function of the data is to be maximized. We want to find the values of the population parameters μ, L and Ψ. As mentioned earlier, the solution for these factor models are not unique. Equivalent models can be obtained by rotation. To obtain a unique solution an additional constraint to be imposed is that $L'\,\Psi^{-1}\,L$ is a diagonal matrix.

Computationally this process is complex. In general, there is no closed- form solution to this maximization problem, so iterative methods must be applied.

Source: Adapted from Applied Multivariate Statistical analysis by the Johnson R.A. and Wichern D.W. (2002).

13.7 FACTOR ROTATION

Broadly speaking, there are two kinds of rotations i.e. orthogonal rotation and oblique rotation. Orthogonal rotation assumes that the factors are uncorrelated. This is less realistic since factors generally are correlated with each other to some extent or the other. Two commonly used orthogonal rotation techniques are Quartimax and Varimax rotation. Quartimax involves the minimization of the number of factors needed to explain each variable whereas Varimax minimizes the number of variables that have high loadings on each factor and works to make small loadings even smaller.

All factor loadings obtained from the initial loadings by an orthogonal transformation have the same ability to reproduce the covariance matrix. From matrix algebra, we know that an orthogonal transformation corresponds to a rigid rotation of the coordinate axis. For this reason, an orthogonal transformation of the factors is called factor rotation.

If $\hat{L}$ is the p×m matrix of estimated factor loadings obtained by any one of the method i.e., principal component or maximum likelihood, then

$$\hat{L}^* = \hat{L}\,T \qquad \text{where} \quad TT' = T'T = I \qquad (13.13)$$

is a p×m matrix of rotated loadings. Moreover, the estimated covariance matrix remains unchanged, since

$$\hat{L}\hat{L}' + \hat{\Psi} = \hat{L}TT'\hat{L} + \hat{\Psi} = \hat{L}^*\hat{L}^{*\prime} + \hat{\Psi} \qquad (13.14)$$

Equation (13.14) shows that the residual matrix, $S_n - \hat{L}\hat{L}' - \hat{\Psi} = S_n - \hat{L}^*\hat{L}^{*\prime} - \hat{\Psi}$, remains unchanged. Moreover, the specific variances $\hat{\Psi}_i$ and hence the communalities $\hat{h}_i^2$ are unaltered. Thus from a mathematical viewpoint, it is immaterial whether $\hat{L}$ or $\hat{L}^*$ is obtained.

The various factor rotation methods have, as a guiding principle, the simple structure concepts. That is to say, the results after rotation should become simple in their appearance. To put it another way, these simple structure concepts should be considered when trying to determine whether or not a given factor rotation has clarified the underlying structure of the data.

Factors are rotated for better interpretation since un-rotated factors are ambiguous. The goal of rotation is to attain an optimal simple structure which attempts to have each variable load on as few factors as possible, but maximizes the number of high loadings on each variable. Ultimately, the simple structure attempts to have each factor define a distinct cluster of interrelated variables so that interpretation becomes easier. For example, variables that relate to language should load highly on language ability factors but should have close to zero loadings on mathematical ability.

Varimax Rotation: Perhaps the most widely used is the varimax criterion. It seeks the rotated loadings that maximize the variance of the squared loadings for each factor; the goal is to make some of these loadings as large as possible, and the rest as small as possible in absolute value. The varimax method encourages the detection of factors each of which is related to few variables. It discourages the detection of factors influencing all variables. The quartimax criterion, on the other hand, seeks to maximize the variance of the squared loadings for each variable, and tends to produce factors with high loadings for all variables.

Kaiser [5] has suggested an analytical measure of simple structure known as the varimax (normal varimax) criteria. Let us define $\tilde{l}_{ij}^{*4} = \hat{l}_{ij}^*/\hat{h}_i$ to be the rotated coefficients scaled by the square root of the communalities. Then the (normal) varimax procedure selects the orthogonal transformation T that makes

$$V = \frac{1}{p}\sum_{j=1}^{m}\left[\sum_{i=1}^{p}\tilde{l}_{ij}^{*4} - \left(\sum_{i=1}^{p}\tilde{l}_{ij}^{*2}\right)^2/p\right] \qquad (13.15)$$

as large as possible.

Scaling the rotated coefficients $\hat{l}_{ij}^*$ has the effect of giving variables with small communalities relatively more weight in the determination of simple structure. After the transformation T is determined, the loadings $\tilde{l}_{ij}^*$ are multiplied by $\hat{h}_i$ so that the original communalities are preserved.

In the above equation, it has a simple interpretation. In words

$$V \propto \sum_{j=1}^{m}\left(\text{variance of the squares of (scaled)loadings for } j^{th} \text{ factor}\right)$$

One can maximize V that corresponds to the squares of the loadings on each factor as much as possible. Therefore, we hope to find groups of large and negligible coefficients in any column of the rotated loadings matrix $\hat{L}^*$.